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Credit Access and Financial Inclusion: An Empirical Study of Indian MSME's

Course Project

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
MSME	Micro, Small, Medium Enterprises
NBFC	Non-Banking Financial co operation
ONDC	Open Network for Digital Commerce
GDP	Gross Domestic Product
CGTMSE	Credit Guarantee Fund Trust for Micro and Small Enterprises

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EXECUTIVE SUMMARY

MSMEs are integral to the Indian economy, significantly contributing to GDP, manufacturing, exports, employment, and local development. Yet the sector faces a severe shortage of affordable formal credit: an estimated debt demand of ₹64 lakh crore against a formal supply of ₹34 lakh crore leaves a ₹30 lakh crore gap.

This technical report investigates the structural causes of the MSME credit gap, its business impacts, and institutional solutions. It combines secondary institutional data (RBI, SIDBI–TransUnion CIBIL, Ministry of MSME) with a primary online survey of 42 MSME owners and workers covering enterprise characteristics and credit experiences.

The findings reveal that credit exclusion extends beyond outright rejections (20%). Only 10% received their full requested amount, 37.5% received reduced funds, and 30% were discouraged by procedural hurdles. Documentation, strict collateral requirements, credit history constraints, and processing delays are major barriers, heavily impacting micro-enterprises with limited fixed assets and irregular cash flows.

Primary Survey data highlights that 95% of respondents face unmet capital requirements, with nearly half needing ₹50,000–₹2 lakh primarily for short-term stock reconstitution. Adequate capital enables market expansion, better margins, capacity addition, and reduced reliance on expensive informal debt. Case comparisons demonstrate that unfinanced enterprises operate at only 40% capacity while borrowing at 24–36% interest, whereas adequately financed peers operate at 85% capacity with higher turnover and employment.

Closing the credit gap requires more than simply boosting financial supply. Key recommendations include simpler documentation, faster processing, reduced collateral dependence, stronger credit guarantees, and broader adoption of cash-flow- and data-based lending via GST records, digital transaction histories, and Account Aggregator infrastructure. Coordinated efforts among government, banks, NBFCs, FinTechs, and MSMEs are essential to make formal credit accessible and timely.

INTRODUCTION

1.1 Background and Motivation

In modern industrial economics, debt is a fundamental engine of operational growth, working-capital liquidity and enterprise risk management. Even the world's most profitable tech conglomerates leverage on huge debt facilities to grow their operations and capture market shares: Alphabet-Google has over \$13 billion in long-term debt. Amazon is managing a long-term debt load of over \$58 billion to sustain its global logistical footprint. In India the established giants use leverage to fund their R&D and manufacturing capabilities like Reliance Industries (over ₹3.1 lakh crore in borrowings) and Tata Motors (over ₹1.2 lakh crore in gross debt).

Institutional lenders underwrite the multi-thousand crores credit lines based on projected cash flows and brand equity. Even witnessed high-profile corporate gambles fail (witnessed in the high-profile debt defaults and wilful bankruptcies of figures like Vijay Mallya and Nirav Modi) but institutional capital flows to large-scale corporate syndicates.

This creates a glaring economic paradox: why wouldn't a small business owner, local apparel manufacturer or farmland processor struggle to get even ₹50,000 to ₹5 lakh in institutional credit to buy raw materials and meet a verified order having a good cash flow genuine business that been trusted by customers for years.

Micro, Small and Medium Enterprises are the operational bedrock of the Indian economy, prime drivers of grassroots entrepreneurship, industrial production and broad-based employment. Represented by an estimated network of over 63 million enterprises:

Generating approx. 30% of India's Gross Domestic Product (GDP). Accounting for nearly 45% of national manufacturing output. Driving over 42% of total merchandise exports. Employing over 110 million individuals, being the second-largest employer after agriculture and prime driver of socio-economic mobility in Tier-2, Tier-3 and rural regions.

Whilst India's macroeconomic indicators show rapid digital modernisation and financial market deepening, this capital expansion has bypassed the informal industrial periphery. These micro-enterprises operate on paper-thin margins, volatile cash flows and are structurally cut-off from formal credit channels, institutional resilience and modern technological upgrades

1.2 The Problem Statement

The main problem with the commercial lending system in India is its imbalances in credit availability for the MSME sector. Even though grassroots-level borrowing needs to fund working capital and inventories the institutional capital can only meet a small proportion of the overall demand, leaving a ₹30 lakh crore

Rs. deficit. By analysing the situation and reviewing the collected data this technical paper will address the following questions:

- What are the reasons why institutional capital fails to meet the bottom-up needs of local enterprises? This paper will review the identified stress points in the current credit distribution mechanism and analyse how they affect the ability of MSMEs to secure loans at any given point.
- What are the effects of such a gap on both businesses and the larger economy (who is affected and how)? This paper will assess the impact of credit unavailability on the loan-seeking businesses and review the indirect effects on local economies.
- What does the government do to address the issue? This paper will evaluate the relevant policy initiatives developed by the Government of India and the Reserve Bank of India.
- What can be done to address the problem? This paper will propose possible interventions and suggest ways for the banking sector to adjust to better meet the needs of local businesses.

1.3 Scope and Research Objectives.

Target Audience

- Policymakers and Regulatory Authorities: Officials within NITI Aayog, the Ministry of MSME, and the Reserve Bank of India (RBI) who design macro-financial frameworks.
- Financial Strategists and Fintech Innovators: Commercial banking heads and digital infrastructure developers focused on alternative credit architectures.
- Academic Researchers: Economists and scholars evaluating financial inclusion bottlenecks and structural market failures in emerging economies.

Research Relevance

With the exact, staggering MSME credit deficit having been quantified and brought to national attention recently, the economic urgency of this issue cannot be overstated. When over 99% of MSMEs (specifically the micro-segment) are locked out of formal credit, it structurally suppresses wage growth, limits grassroots industrial output, and forces local entrepreneurs into usurious debt traps. Resolving this capital bottleneck is a critical prerequisite for sustaining India's macroeconomic growth, boosting global export competitiveness, and ensuring equitable socio-economic mobility.

The scope of this project is explicitly focused on dissecting the operational constraints governing micro and small enterprise (MSE) credit access, rather than broad macroeconomic theory. The core research focus encompasses:

- Quantifying the Structural Disconnect: Mapping how traditional, asset-heavy banking models systematically fail to serve cash-reliant, informal local businesses.
- Capturing Ground-Level Realities: Executing and analysing a primary field survey to document real-world loan rejection rates, hidden collateral demands, and the daily dependency on informal moneylenders.

- Evaluating Implementation Friction: Assessing why nominally collateral-free state interventions (such as MUDRA and CGTMSE) often stall at the commercial branch level due to localized risk aversion and bureaucratic paperwork.
- Formulating Scalable Solutions: Investigating the viability of transitioning from asset-backed lending to digital, cash-flow-based underwriting—specifically utilizing India's Digital Public Infrastructure (DPI) like the Account Aggregator (AA) network and GST transaction histories.

METHODOLOGY

2.1 Literature Survey

The structural constraints and empirical realities of the Indian MSME credit deficit have been extensively quantified across a chronological series of regulatory frameworks, institutional audits, and independent economic studies. This section reviews the seminal reports that define the current understanding of the credit gap, its root causes, and proposed systemic solutions.

- **Quantifying the Gap (2018)** The foundational baseline for understanding the scale of this issue was established in 2018 when the *International Finance Corporation (IFC)* published its comprehensive assessment on financing India's MSMEs. This global multilateral study highlighted that the Indian MSME funding void represents a classic market failure of "*missing middle*" finance. Building upon this historical baseline, contemporary data—as specifically detailed in the "Credit gap in MSME" project document—confirms the exact, staggering size of the current unaddressed credit deficit.
- **Identifying the Root Causes (2019):** To investigate why this gap persists despite macroeconomic growth, the RBI convened an expert panel, culminating in the 2019 *Report of the Expert Committee on Micro, Small and Medium Enterprises*, chaired by U.K. Sinha. This landmark regulatory report explicitly identified deep information asymmetry, a widespread lack of enterprise formalization, and the banking sector's rigid reliance on physical collateral mandates as the primary bottlenecks choking capital flow to grassroots businesses.
- **Tracking Ground-Level Realities (2023):** Empirical evidence of lender behaviours and the socioeconomic impact on borrowers is continuously documented in the *MSME Pulse reports*, jointly published by the Small Industries Development Bank of India (SIDBI) and TransUnion CIBIL. The 2023 iteration of this report provides critical data showing that while aggregate commercial lending volumes have expanded, formal capital remains disproportionately concentrated among well-established medium and small enterprises. Consequently, most micro-units are left highly dependent on informal, high-cost lending networks.
- **Evaluating Government Formalization Efforts (2023):** The *Ministry of Micro, Small & Medium Enterprises Annual Report 2022-23* outlines the sheer demographic scale of the sector and tracks the government's ongoing formalization drive via the Udyam registration portal. However, these government white papers frankly acknowledge a persistent structural lag between the rapid digital onboarding of enterprises onto formal registries and the actual, often delayed, disbursement of institutional credit through state-backed schemes.
- **Proposing Digital Solutions (2020–2021):** A critical segment of recent literature focuses on technological infrastructure as the ultimate remedy for traditional underwriting failures. NITI Aayog's 2021 strategy paper on financial inclusion emphasizes the macroeconomic urgency of transitioning toward digital, data-driven credit underwriting to bring Tier-2 and Tier-3 enterprises into the formal banking fold. Furthermore, *technical white papers on the Open Credit Enablement Network (OCEN)* published by CredAll/iSPIRT in 2020, and the Account Aggregator (AA) framework detailed by Sahamati in 2021, advocate for a complete structural paradigm shift. These frameworks propose replacing traditional asset-backed lending with cash-flow-based underwriting, utilizing alternative digital footprints such as Goods and Services Tax (GST) invoices and digital payment histories to assess risk.

2.2 Data Collection: Primary Survey Design & Secondary data

Primary Survey Design

The primary survey was administered via a structured questionnaire to micro- and small-scale business owners to capture ground-level operational and credit realities.

Sampling Methodology: A snowball sampling methodology was adopted, utilizing initial contacts from family and peer networks to sequentially recruit additional qualified participants. This chain-referral approach effectively facilitated data collection across the target business population.

Primary survey architecture (16 Questions)

Section 1	Section 2	Section 3	Section 4
Company Information	Financing Sources	Credit Experience	Unmet Capital issue

Table 2.1: Section in survey

Section 1: Company Information (Questions 1 to 4)

- **Q1: Enterprise Sector & Core Activity:** Categorizes the enterprise into Agro-processing, Manufacturing/Fabrication, Wholesale, Retail, or Services.
 - Credit demand patterns and asset-turnover cycles differ fundamentally between fixed-asset manufacturing and fast-moving retail trade.
- **Q2: Supply Chain Position & Operating Model:** Captures upstream component suppliers, finished goods makers, B2B distributors, B2C retail, and hybrid models.
 - Identifies where cash gets trapped in supply chains (e.g., component makers suffering buyer payment delays vs. cash-and-carry retail).
- **Q3: Total Fixed Capital & Asset Base:** Tiers businesses from Micro Tier I (< ₹1 Lakh) to Medium Scale (> ₹1 Crore).
 - Determines the borrower's mortgageable asset base to evaluate whether lenders mandate collateral disproportionate to the enterprise's asset size.
- **Q4: Monthly Cash Inflows & Stability Pattern:** Evaluates revenue stability from volatile/seasonal flows to high-volume banked receipts.
 - Contrasts cash-dominant businesses against formal-invoicing businesses to assess readiness for digital cash-flow underwriting (e.g., Account Aggregator framework).

Section 2: Financing Sources (Questions 5 to 7)

- **Q5: Working Capital Shocks & Business Stressors:** Measures exposure to 45+ day delayed payments, raw material inflation, footfall decline, family emergency withdrawals, or regulatory disruptions.
 - Identifies the external triggers that push a solvent micro-business into an acute liquidity crisis.
- **Q6: Preference for Source of Operation & Emergency Credit:** Ranks preferences across formal banks, NBFCs/MFIs, supplier trade credit, informal high-cost moneylenders, zero-cost family/chit funds, and retained profits.
 - Maps borrower capital preferences to reveal the operational substitution hierarchy when institutional banks reject applications.

- **Q7: Received Government Subsidies/Loans:** Binary indicator (Yes/No) on state support access.
 - Measures ground-level penetration of flagship public credit programs (PMMY, CGTMSE, PMEGP).

Section 3: Credit Experience (Questions 8 to 12)

- **Q8: Formal Loan Application History & Outcome:** Categorizes outcomes into Full Approval, Rationed Amount, Formal Rejection, Discouraged Non-Borrower, or Voluntary Non-Borrower.
 - Quantifies both explicit institutional rejection and implicit discouragement (borrowers avoiding banks due to procedural intimidation).
- **Q9: Effective Annualized Cost of Borrowing:** Ranges from prime rates (< 9.5%), standard commercial (10–16%), NBFC micro-loans (17–28%), informal moneylenders (30–60%), to daily meter interest (> 60% p.a.).
 - Measures the high-interest premium micro-enterprises pay in informal markets relative to formal corporate borrowing rates.
- **Q10: Collateral & Guarantee Requirements Enforced:** Identifies security demands including clear-title real estate, gold/machinery, financial deposits, third-party personal guarantees, unsecured cash-flow lending, or loan denial due to collateral absence.
 - Evaluates compliance with RBI guidelines prohibiting collateral mandates on micro-ticket loans under state guarantee thresholds.
- **Q11: Turnaround Time (TAT) & Transaction Costs:** Spans Instant/48 hours (informal/fintech), 3–7 days (SFBs/gold loans), 2–4 weeks (private banks), 1–3 months (PSU banks), to application abandonment.
 - Documents operational latency; extended banking turnaround times often render approved loans useless for urgent inventory purchase cycles.
- **Q12: Structural Friction Points in Accessing Formal Finance:** Isolates specific bottlenecks: lack of formal GST/audited ITRs, property valuation/title disputes, thin credit file, loss of business days from repeated branch visits, and agent rent seeking/broker commissions.
 - Identifies exact operational barriers to inform targeted digital public infrastructure and policy solutions.

Section 4: Unmet Capital Deficit (Questions 13 to 16)

- **Q13: Current Unmet Capital Deficit (The Immediate "Credit Gap"):** Tiers capital needs from Zero gap, Micro gap (₹50,000–₹2,00,000), Small gap (₹2,00,001–₹10,00,000), Substantial gap (₹10,00,001–₹50,00,000), to Strategic expansion (> ₹50 Lakh).
 - Provides direct primary quantification of the unaddressed liquidity requirement per enterprise unit.
- **Q14: Growth Trajectory Impact of Bridging the Gap:** Identifies tangible economic outcomes: bulk cash supplier discounts (10–20%), capacity/machinery scaling, geographic expansion, replacing toxic high-interest debt, or extending customer credit terms.
 - Quantifies the opportunity cost of credit exclusion on enterprise profitability and scaling potential.
- **Q15 & Q16: Qualitative Open-Ended Responses:**
 - *Q15:* Captures narrative descriptions of bank branch interactions, documentation hurdles, and informal rejections.

- *Q16*: Documents specific business losses, cancelled orders, and stunted opportunities directly resulting from credit denial.

Secondary Data Sets

To establish a macro-level baseline, four key secondary institutional datasets were selected. Each serves a distinct analytical purpose in quantifying the credit shortfall and assessing systemic risk:

- **Reserve Bank of India (RBI) – Database on Indian Economy (DBIE):**
 - Longitudinal data on Priority Sector Lending (PSL), sectoral deployment of gross bank credit, and Non-Performing Asset (NPA) ratios segmented by industry scale.
 - Provides empirical evidence of commercial banking exposure to Micro vs. Small vs. Medium enterprises, allowing direct comparison of corporate credit volume against priority micro-lending.
- **Ministry of MSME – Udyam Registration Portal & Dashboard:**
 - *Key Features*: Granular demographic records of formalized enterprises, state-wise geographic density, sector categorization (manufacturing vs. services), and gender/social category of enterprise owners.
 - Establishes the total universe of formalizing micro-units and maps the structural skew where over 99% of registered entities belong strictly to the Micro tier.
- **SIDBI – TransUnion CIBIL (*MSME Pulse*):**
 - *Key Features*: Credit commercial bureau data tracking loan origination rates, delinquency trends, credit-seeking behaviour, and commercial score distribution (CMR ratings) across ticket sizes.
 - Dissects commercial lender risk appetite, proving that formal lending expansion remains concentrated in upper-ticket tiers while micro-enterprises face credit rationing.

2.3 Case Study Selection & Survey Comparative Framework

Comparative research design to examine the role of credit access in the financial and operational performance of Micro, Small, and Medium Enterprises (MSMEs). The analysis combines macro-level MSME indicators with enterprise-level case comparisons to assess the relationship between access to formal credit, production capacity, revenue generation, and employment. The methodology is structured into four stages: macroeconomic and MSME sector analysis, credit-gap assessment, enterprise-level case study analysis, and comparative visualization of credit-access outcomes.

Secondary data relating to the Indian MSME sector were compiled from government publications, financial-sector reports, and MSME-related databases. The variables considered include the number of MSMEs, employment generation, contribution to GDP and manufacturing output, export contribution, ownership characteristics, enterprise size, gender composition, formal and informal credit usage, and sources of finance.

The analysis also incorporates information on aggregate debt demand, formal credit supply, and the estimated credit gap to evaluate the extent of unmet financing requirements among MSMEs.

For the enterprise-level analysis, three representative cases were constructed to represent different levels of credit accessibility:

- Case A – Micro enterprise: No formal credit access
- Case B – Small enterprise: Partial formal credit access
- Case C – Medium enterprise: Adequate formal credit access

The cases are compared using production capacity, capacity utilization, annual revenue, formal and informal credit, direct employment, indirect employment, and employment type. The first stage evaluates the overall economic significance and structural characteristics of the MSME sector. Key indicators such as the number of enterprises, employment, GDP contribution, manufacturing contribution, and exports are presented using descriptive statistics.

For Overall Understanding

The MSME population is further classified according to:

- Enterprise size: Micro, Small, and Medium
- Ownership category
- Gender of ownership
- Formal and informal enterprise type
- Sectoral distribution of employment

Percentages and aggregate values are used to identify the dominant characteristics of the MSME sector and establish the economic context for the subsequent credit-access analysis. The second stage evaluates the financing constraints faced by MSMEs. Aggregate addressable debt demand is compared with the estimated formal credit supply to determine the financing gap.

The credit gap is calculated conceptually as:

$$\text{Credit Gap} = \text{Addressable Debt Demand} - \text{Formal Credit Supply}$$

The analysis further examines variations in the credit gap across:

- Enterprise size
- Business sector
- Geographical region
- Gender of ownership

The resulting values are presented graphically to identify groups experiencing relatively higher financing deficits. This provides a sectoral and structural view of the unmet credit requirements of MSMEs. The study examines the sources of financing available to MSMEs by distinguishing between formal credit, informal credit, and enterprises with no credit access. Formal financing is further classified according to the type of financial institution, including:

- Private-sector banks
- Public-sector banks
- NBFCs/FinTech platforms
- Other financial sources

The proportions are compared across micro, small, medium, and overall MSME categories. This analysis is used to identify differences in financing patterns across enterprise sizes and to assess the role of NBFCs and FinTech platforms in expanding alternative credit access.

To examine the practical effect of credit availability, three representative MSME cases are compared. The cases are designed to reflect increasing levels of formal credit accessibility while maintaining differences in enterprise scale.

The following variables are considered: of 3 businesses

- Enterprise size
- Credit access
- Annual capacity
- Capacity utilization
- Annual revenue
- Formal credit

The cases are compared to assess whether greater access to formal finance is associated with higher capacity utilization, production, revenue, and employment generation. The study also examines the relationship between credit availability and financial performance through a revenue-growth trajectory. Revenue growth rates are plotted across five years to visualize changes in financial performance over time. The trajectory is used as a descriptive indicator rather than as evidence of a statistically causal relationship. The purpose is to demonstrate how improved access to working capital and formal finance can support enterprise expansion and revenue generation.

Visualization

- Gauge charts for major MSME macroeconomic indicators
- Bar charts for ownership, size, gender, and financing patterns and Survey answers.
- Donut charts for sectoral employment distribution and Survey answers.
- Credit-gap comparison charts
- Grouped bar charts for production comparison
- Line/area charts for revenue-growth trajectories
- Proportional area Chart: For survey Results

The framework assumes that restricted access to formal finance can constrain working capital and investment, which may subsequently reduce production capacity utilization and limit revenue and employment generation. The case-study and descriptive analyses are therefore used to examine this relationship at both the sectoral and enterprise levels.

RESULTS

3.1 Empirical Data Analysis

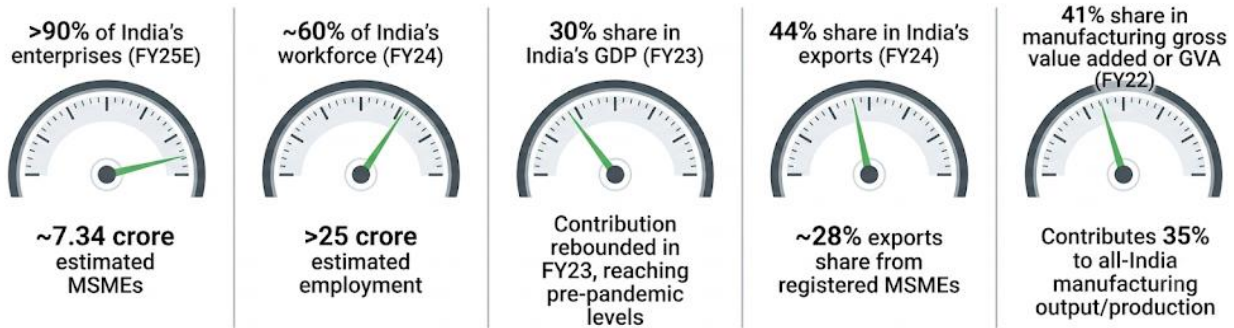


Figure 3.1.1: Macro Indicators (DBIE report)

The Micro, Small, and Medium Enterprises (MSME) sector is one of the most important pillars of the Indian economy. As of now, the sector consists of about 7.34 crore registered entities. The sector is one of the significant contributors to the country's employment rate as it provides jobs to more than 25 crore people and contributes to around 60% of the total employment rate. In terms of the Gross Domestic Product (GDP), the MSME contributes to about 30% of the GDP, 35% of the country's manufacturing production, and 44% of the total exports of the country. Despite the relevance of this sector to the country, most of the enterprises that fall under this category are micro and informal. According to the sizes, it was reported that about 63% of MSMEs are small enterprises, whereas medium enterprises contribute to about 10% of the total enterprises. As per the ownership, it was reported that about 59% of the enterprises are proprietorships. Besides, the report also revealed that male-owned enterprises account for about 88% of the total MSMEs.

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Period	Total Employment (Crore)	Registered MSMEs (Crore)
Cumulative (Mar 2025)	25.95	6.2

Table 3.1.1: Employment Generation (MSME pulse)

Credit Access Challenges

While the sector contributes significantly to the economy, most micro, small and medium enterprises (MSMEs) work with limited funds. Various surveys conducted with enterprise owners have revealed that the most common constraint for them is the limited access to credit. In fact, this challenge was reported by 22% of the respondents, while competition for markets came in second with 18%, followed by issues with technology (10%) and regulations (8%). The numbers reflect the true scenario, as on an average, the MSME sector requires around ₹123 lakh crore to meet its total financial needs, out of which ₹92 lakh crore is the debt component. Taking into consideration only the short-term or working capital needs, which constitutes to around 70% of the debt requirements, the demand for credit rises to ₹64 lakh crore. However, the ability of the financial system, including banks and non-banking financial companies (NBFCs), to meet this requirement is at ₹34 lakh crore. This leaves a gap in credit availability at ₹30 lakh crore.

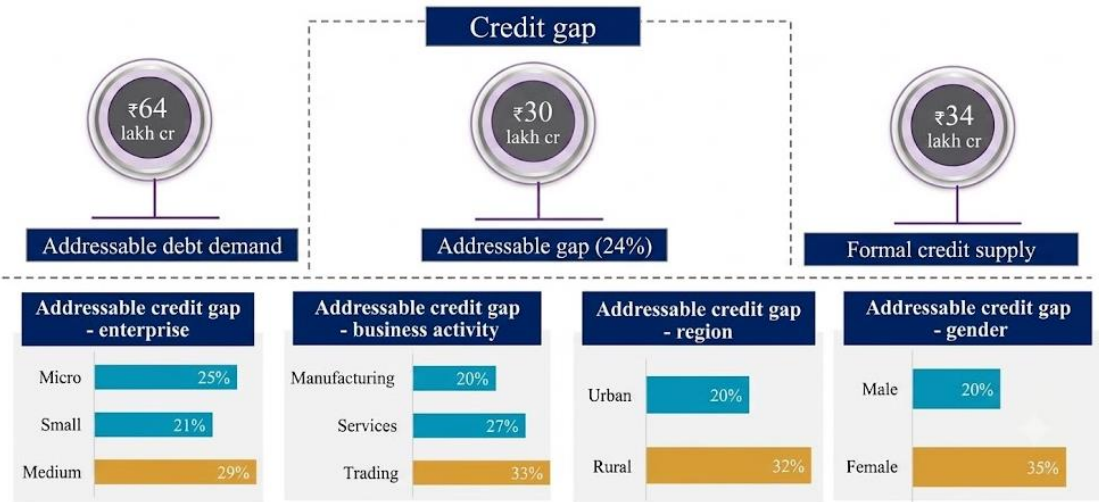


Figure 3.1.2: Credit gap analysis and formal lending deficit (MSME Pulse)

Government Initiatives

The Government of India and the Reserve Bank of India have tried to narrow the MSME credit gap through several measures that work on registration, guarantees, payments and digital lending. The Udyam registration portal gives enterprises a formal identity, and registrations on Udyam and the Udyam Assist platform together crossed 8.7 crore by June 2026. To reduce the collateral barrier, the Credit Guarantee Fund Trust for Micro and Small Enterprises (CGTMSE) offers guarantee cover on collateral-free loans. Its cover ceiling was raised from ₹5 crore to ₹10 crore, and by mid-2025 it had backed over 1.18 crore guarantees worth ₹9.80 lakh crore, including a record ₹3 lakh crore in FY25 alone. RBI has also directed banks not to demand collateral on small micro and small enterprise loans and has set up a revival and rehabilitation framework for stressed MSME accounts. Schemes such as MUDRA and PMEGP target the smallest borrowers, while the TReDS platform and the MSME Samadhaan portal address delayed buyer payments, which our survey identified as a cash-flow stress for 20% of respondents. On the technology side, the Account Aggregator framework, the Open Credit Enablement Network (OCEN) and GST data are meant to let lenders assess repayment ability from cash flow rather than property. Even so, the credit gap of about ₹30 lakh crore persists, which suggests that the problem lies more in implementation than in the design of schemes.

Credit Access Patterns by Enterprise Characteristics

Primary research conducted among MSME proprietors unveiled diverse credit access trends. Of the respondents, 17% did not use credit, while 8% only accessed informal credit. Micro enterprises were more likely to use informal credit than small and medium enterprises (3% and 2% respectively). The percentage of micro enterprises that accessed informal credit was 12%. Gender differentiated the likelihood to obtain credit. Female led MSMEs had a credit offtake of 76% compared to 84% for their male counterparts. 24% of the female headed enterprises did not utilise any form of credit. Moreover, female entrepreneurs were more likely to utilise informal credit than their male counterparts.

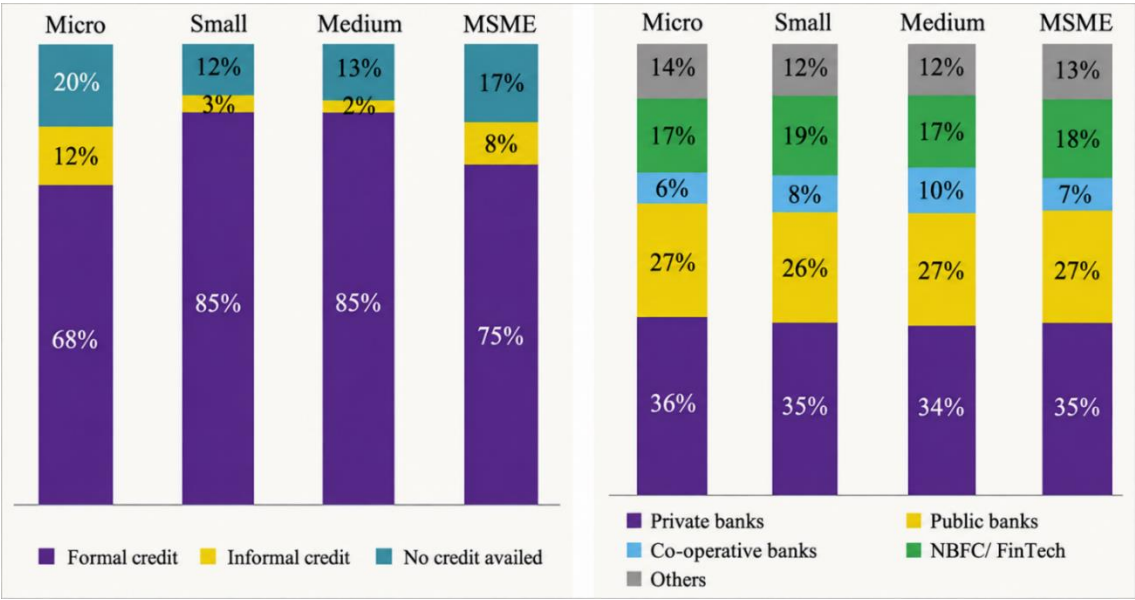


Figure 3.1.3: Financing sources (SIDBI report)

NBFC and Fintech Participation

According to the data on the lending landscape, there is an upward trend in credit supply from non-banking financial companies (NBFCs) and financial technology (fintech). In particular, the provision of credit via alternative lenders has grown steadily during the past years. At the same time, credit supply to the MSME sector has also grown. Nevertheless, according to quantitative analysis, credit demand remains unfulfilled, which implies that credit supply growth has not caught up with credit demand.

3.2 Primary Survey Findings

Our primary data were collected through Online surveys given to 42 MSME owners and workers, who were based mainly in Vellore and Kozhikode regions. We ran our online survey campaign for 48 days. Since our sampling was confined to these two regions, we do not claim to have gathered comprehensive evidence representative of the larger population. Therefore, in our analysis, we focus on general observations made across respondents and do not attempt to extrapolate our findings to make inferences on the larger population. The survey comprised four sections: enterprise description, financing sources, credit experience, and unmet capital needs.

Section 1: Enterprise Profile

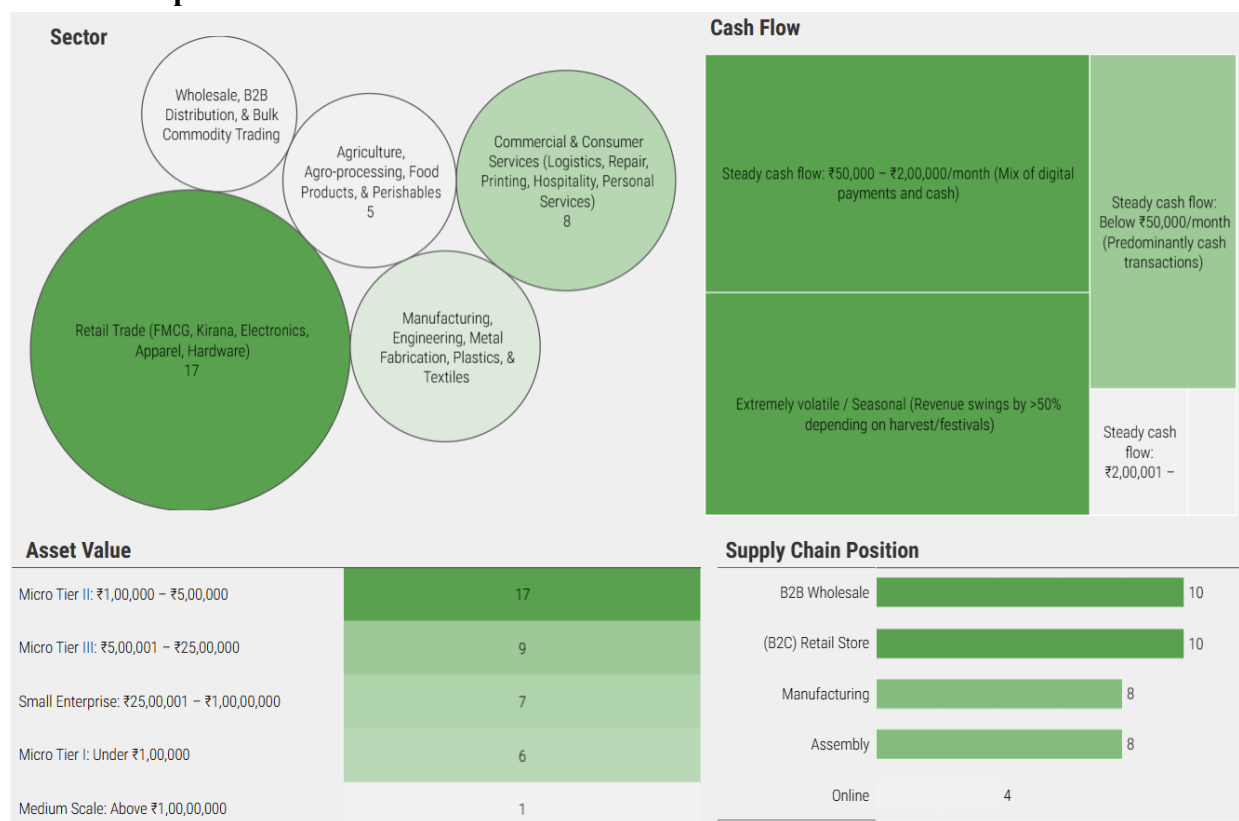


Figure 3.2.1: sector, cash flow, asset value, and supply chain position

The sample was skewed towards respondents engaged in retail trade, which comprised 17 respondents (42.5%). This was followed by those in commercial and consumer services (8 respondents, 20.0%) and agro-processing and food products (5 respondents, 12.5%). Manufacturing and textiles and wholesale and bulk trading were each represented by 6 respondents (15.0%) and 4 respondents (10.0%), respectively. Since the respondents were involved in different types of business, their credit requirements were also different. Whereas retail traders sought credit to replenish their stocks, the manufacturers requested loans to finance the purchase of machinery and raw materials.

The supply chain positions of the respondents were also diverse. Thus, B2B wholesale and direct-to-consumer retail trade positions were each occupied by 10 respondents (25.0%). In turn, raw material processing and finished goods assembly were each adopted by 8 respondents (20.0%). Hybrid and online business models were each selected by 4 respondents (10.0%). The position in the supply chain also influences the extent to which a firm is reliant on credit. For instance, companies crediting other businesses are more likely to experience delays in collecting payments from their debtors, compared to retailers who must ensure a constant supply of goods.

The asset bases of the respondents varied between large and small enterprises. Thus, 17 respondents (42.5%) possessed between ₹1-5 lakh, whereas 9 respondents (22.5%) reported asset levels between ₹5-25 lakh. In turn, 6 respondents (15.0%) had below ₹1 lakh, and only 7 respondents (17.5%) were small-scale enterprises worth between ₹25 lakh to ₹1 crore. Finally, one respondent indicated that their enterprise was worth more than ₹1 crore. Overall, 32 respondents (80%) belonged to the bottom three categories.

Since a significant proportion of the respondents were small enterprises, their asset base might be inadequate to secure a loan against.

The cash flow positions of the respondents were also disparate. Thus, many respondents indicated the presence of seasonal or highly variable cash flows spanning above 50% (14 respondents, 35%) or steady cash flows, with monthly amounts between ₹50,000 and ₹2 lakh (12 respondents, 30%). A stable cash flow between 0 and ₹50,000 was reported by 9 respondents (22.5%). Only one respondent (2.5%) indicated that their monthly cash flows exceeded ₹2 lakh. Many respondents, therefore, especially those with highly variable cash flows, may find it challenging to provide the lender with the required documentation to demonstrate their ability to repay the loan.

Section 2: Financing Sources and Business Stressors



Figure 3.2.2: stress factors and government subsidy

Increasing input costs which include raw material, transportation, and energy costs were cited by 17 respondents (42.5%) as the major financial shock. In the second position, demand shocks were cited by 9 respondents (22.5%). Delayed payments from customers of 45 days or more were cited by 8 respondents (20%). Three respondents (7.5%) cited the need to liquidate business funds to meet personal financial commitments. Two respondents (5%) each cited regulatory or disruption shock as a financial shock experience. One respondent (2.5%) noted that he/she had not experienced any of these shocks. It can be gathered from the results that business owners usually borrow funds to meet working capital needs rather than long-term investment needs.

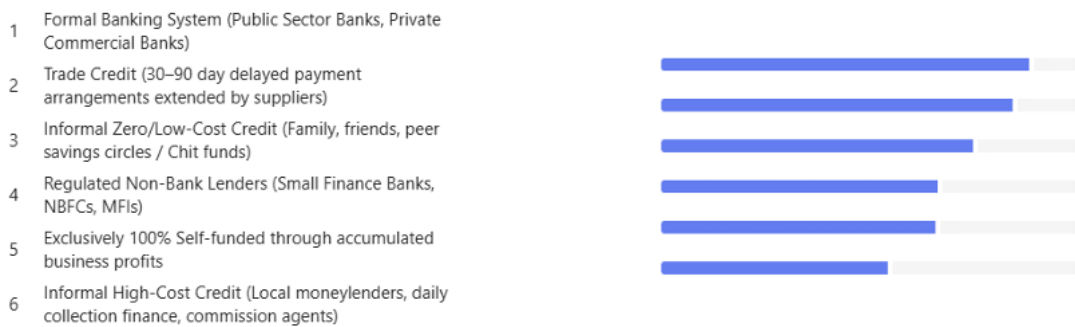


Figure 3.2.2.1: preferred credit source

On preferred sources of emergency finance, respondents ranked formal banks first and supplier trade credit second. Informal low-cost sources such as family, friends, and savings circles ranked third, followed by regulated non-bank lenders (small finance banks, NBFCs, and MFIs). Self-funding from profits ranked fifth, and high-cost moneylenders ranked last. Informal high-cost borrowing therefore does not appear to be a first choice. It is more likely a fallback when preferred sources fall short.

Government support had limited reach in the sample. Only 12 respondents (30.0%) had ever received a subsidy or government-backed loan, while 28 (70.0%) had not.

Section 3: Credit Experience

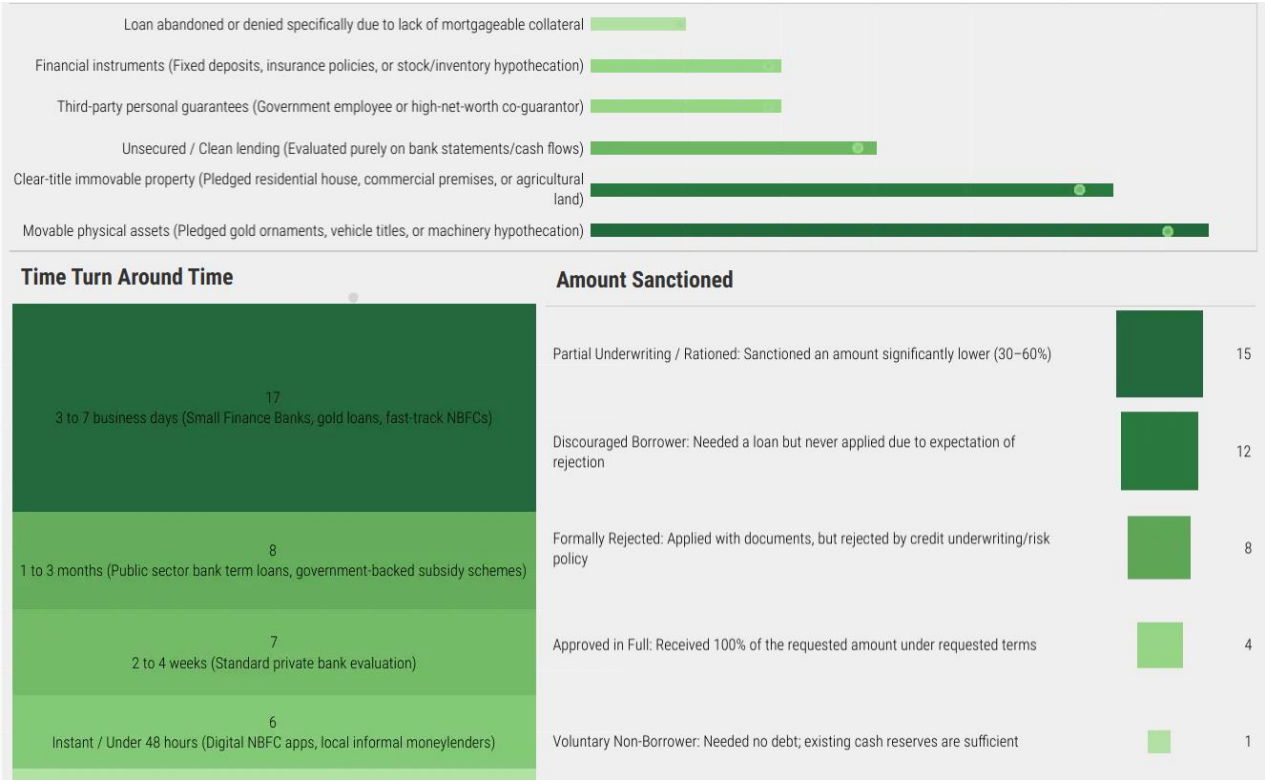


Figure 3.2.3: reasons for rejection, Challenges

Only 4 respondents (10.0%) received the full amount they asked for. The largest group, 15 (37.5%), had their loans scaled down, receiving significantly less than requested. Eight (20.0%) were formally rejected, and 12 (30.0%) were discouraged borrowers who needed credit but did not apply because they expected rejection or a heavy process. One respondent (2.5%) had no need for credit. Counting only formal rejections would understate the problem, since constraints appear before and during the application as well as at the decision stage.

Respondents named several barriers. Documentation was the most common at 12 respondents (30.0%), covering GST filings, audited accounts, and three years of tax returns. Collateral problems were next at 8 (20.0%), followed by thin credit history or past defaults at 7 (17.5%), and process friction from repeated branch visits and complex forms at 7 (17.5%). Six respondents (15.0%) reported having to pay brokers or agents upfront. No single barrier explains exclusion; they overlap.

Security demands were widespread. Lenders most often asked for movable assets such as gold, vehicles, or machinery, and for clear-title property. Financial instruments and third-party guarantees were asked for less often, and a smaller group received clean lending based on bank statements. A few respondents abandoned or lost a loan solely for lack of mortgageable collateral. Borrowing costs varied widely. Most respondents who borrowed paid 10–16% a year on standard formal credit, a sizeable group paid 17–28% on higher-risk formal or NBFC credit, and a few faced informal rates of 30% or more. A small number accessed subsidized rates below 9.5%.

Processing time also carried a cost. The most frequent turnaround was 3–7 business days, reported by 17 respondents (42.5%), mainly from small finance banks, gold loans, and fast NBFCs. Six (15.0%) were funded within 48 hours, mostly through digital NBFC apps or local moneylenders. Seven (17.5%) waited 2–4 weeks, and 8 (20.0%) waited one to three months, typically for public-sector bank or subsidy-linked loans. Two respondents (5.0%) gave up because of documentation and inspection delays. The fastest sources were often the most expensive, while cheaper institutional credit took longer.

Section 4: Unmet Capital Needs

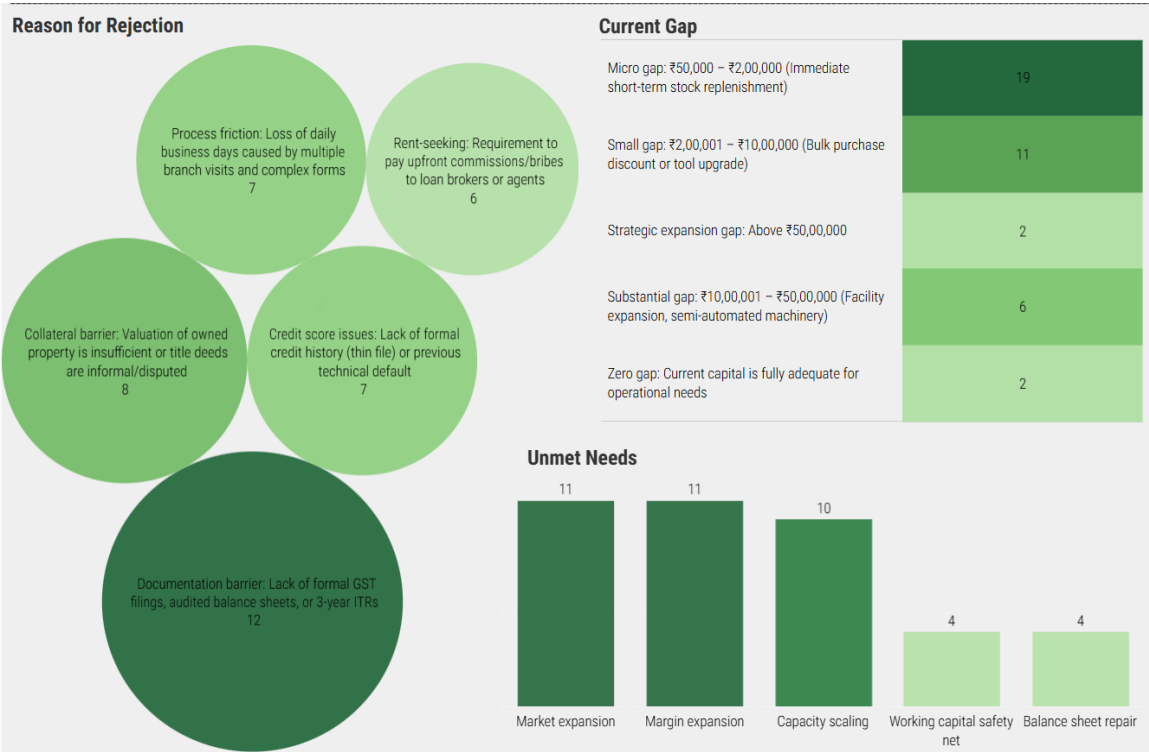


Figure 3.2.4: current gap and unmet needs

Thirty-eight of 42 respondents (95.0%) reported an unmet capital need. Only 2 (5.0%) said their current capital was fully adequate. The largest group, 19 respondents (47.5%), needed ₹50,000 to ₹2 lakh, mainly for short-term restocking. Another 11 (27.5%) needed ₹2 to ₹10 lakh for bulk purchases or tool upgrades, 6 (15.0%) needed ₹10 to ₹50 lakh for expansion or semi-automated machinery, and 2 (5.0%) needed over ₹50 lakh. Most gaps are small in absolute terms, which points to a shortfall in working capital and modest expansion funds rather than large project finance.

Asked what they would do with sufficient capital, 11 respondents (27.5%) chose market expansion and another 11 (27.5%) chose margin expansion, such as buying in bulk at supplier discounts. Ten (25.0%) would add capacity through machinery, production lines, or storage. Four (10.0%) would replace expensive informal debt, and 4 (10.0%) would offer longer credit terms to business buyers. The unmet need therefore constrains margins, capacity, and growth as well as day-to-day cash.

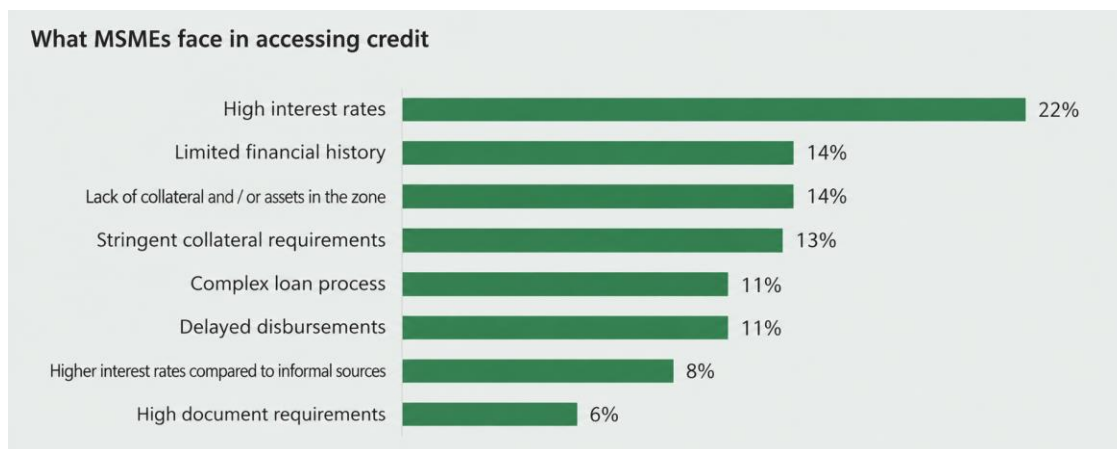


Figure 3.2.5: keyword in response

A few open-ended responses support these patterns. Respondents mentioned strict documentation, the difficulty of proving a low-income business without GST-verified records, repeated meetings without a clear answer, and expansion plans such as a second shop that never went ahead for lack of capital. Since only a handful answered, these are treated as illustrations rather than evidence.

3.3 Case Study Analysis

Indicator	Case A: Micro complementary unit (no formal credit)	Case B: Small jewellery unit (partial credit)	Case C: Bags and apparel production unit (adequate credit)
Installed annual capacity (pieces)	60,000	10,00,000	18,00,000
Capacity utilization	38–42%	65–70%	82–88%
Annual turnover	₹6–9 lakh	₹30–38 lakh	₹90 lakh – ₹1.1 crore
Formal credit	None	₹10–15 lakh (NBFC, 16–18% p.a.)	₹42–50 lakh (bank loans, 10–12% p.a.)
Informal credit	Regular reliance (24–36% p.a.)	Occasional, limited	None
Direct jobs	2–3	8–12	30–42
Indirect jobs (home-based, job-work, suppliers)	3–5	15–20	60–80
Nature of employment	Casual, piece-rate	Mixed	Mostly permanent, salaried

Table 3.3.1: Comparison case study

We compared three Indian MSMEs in the accessories and light-manufacturing space to show how access to finance relates to business performance and livelihood outcomes. The comparison covers capacity utilization, job creation, revenue, borrowing costs, and quality of work. The units are a micro enterprise making complementary jewellery items such as beaded add-ons, pouches, and display packaging (Case A), a small jewellery manufacturer (Case B), and a bags and apparel production unit (Case C). Together they span a credit spectrum from no formal credit to adequate institutional credit.

Units in this segment typically face tight working capital for buying metals, beads, fabric, and trims, along with limited funds for machinery such as stitching and polishing equipment. Demand is also seasonal, peaking around festivals and weddings, so firms without a cash buffer risk missing bulk orders.

Case A uses roughly two-fifths of its capacity. With no formal lender, the owner borrows from moneylenders and chit arrangements at 24–36% a year, which limits how much raw material can be bought at once. Turnover is ₹6–9 lakh, supporting 2–3 casual, piece-rate jobs.

Case B uses an NBFC loan to run at about two-thirds of capacity and earn ₹30–38 lakh a year, supporting 8–12 direct and 15–20 indirect jobs. Interest of 16–18% eats into its margins, however, and slows growth.

Case C draws on bank finance at 10–12% to fund machinery and bulk fabric purchases. It runs at about 85% of capacity, turns over around ₹1 crore, and employs 30–42 people directly and 60–80 indirectly, mostly in permanent roles.

Capacity utilization. Higher credit access goes with fuller use of capacity. Using midpoints (42% for Case A, 85% for Case C), the gap is about 45 percentage points. The well-financed unit can buy inputs ahead of demand, while the credit-starved unit stops when cash runs out.

Employment generation. Using midpoints, direct jobs rise from 2.5 to 35 (about 14 times) and indirect jobs from 4 to 70 (about 17 times). Part of this reflects Case C's larger scale and different trade, so the multiples are indicative rather than purely credit driven.

Quality of employment. Casual piece-rate work in Case A gives way to a mixed model in Case B and salaried permanent staff in Case C, suggesting that predictable cash flow supports steadier employment.

Revenue and local economy. Midpoint turnover rises from about ₹7.5 lakh to about ₹1 crore, roughly 13 times. This likely raises incomes for workers, artisans, and local suppliers, though these wider effects were not measured.

Financial sustainability. Borrowing costs fall from 24–36% (Case A) to 16–18% (Case B) to 10–12% (Case C). Costlier finance leaves less to reinvest, which probably makes it harder for smaller units to build reserves and expand. This is an inference from the pattern; a direct causal link was not tested.

Overall assessment. The cases suggest that affordable institutional credit is closely associated with better socio-economic outcomes. The unit without formal credit shows low-capacity use, few unstable jobs, expensive borrowing, and small turnover. The adequately financed unit shows high-capacity use, larger and more secure employment, cheaper finance, and far higher turnover.

DISCUSSION

4.1 What the Three Sources Say Together

The national data speaks of a sector that carries a great deal of weight but is thinly financed. MSMEs account for roughly 30% of GDP and around 60% of employment, but the formal system provides only ₹34 lakh crore against an addressable demand of ₹64 lakh crore, a shortfall of about ₹30 lakh crore. Our survey shows what that shortfall looks like from the owner's perspective and case comparison shows what it means for output and employment. When the three are laid side by side they are pointing in the same direction.

The first finding that stands out is that it is not a simple matter of being turned down. Only 8 of our 42 respondents were formally rejected but another 15 had their loan amounts cut well below the requested figures and 12 were never inclined to apply because of expectations of a "no" or dread of the process. Adding these up and only 4 people got what they wanted. That matches the national data where even the formally served groups get credit mostly from banks and NBFCs and the unmet demand persists. Looking at mere rejections would make the credit gap appear smaller than it is as much exclusion comes quietly before an application is filed or after the amount has been trimmed.

The second finding is that cost and delay seem to matter as much as access. In the secondary survey, high interest rates were the most cited barrier at 22%, well ahead of limited financial history and lack of nearby banks (14% each), collateral (13%) and complex loan processes and delayed disbursements (11% each). The respondents that we interviewed reported a similar story but in a different way. The fastest money, from digital NBFC apps and local moneylenders, was also the most expensive and cheaper public-sector loans could take up to one to three months. A shop that has to stock in time for a festival cannot afford that wait and so ends up borrowing from the sources it ranked lowest in the survey. For this reason, informal high-cost financing appears as less of a preferred option and more of a fallback.

The case comparison adds a third dimension, it has to be read carefully. Case A, with no formal credit and borrowing at 24-36%, ran at about 42% of its capacity. Case C, which is financed by a bank at 10-12%, ran at about 85% and supported far more permanent jobs. We are tempted to lay the credit gap at the root of this difference but the three units also differ in trade and in scale and the safer reading would be that the cases are consistent with the survey rather than proof of it. The survey respondents themselves related unmet capital to specific losses: lower margins due to the absence of bulk-purchase discounts, no room to add machinery and expansion plans for a second shop that never came off.

We have to be honest about the limits of this evidence too. The primary sample is small, drawn by convenience from two regions and consists mostly of micro enterprises. The findings describe the experiences of those 42 businesses and the national figures derive from other studies. What we are able to say is that the patterns are consistent: constraints begin early, are layered and fall hardest on the smallest and least documented firms.

4.2 Why the Gap Persists

If the issue was simply one scarcity of credit, more of it would do the trick. The situation is more complicated than that, and the national sample indicates the same. The credit supply by non-banking financial institutions and fintech platforms has been growing steadily, but the demand has not dried out in tandem. Our survey helps to explain that, too: documentation was the largest issue, which 12 respondents cited as the reason for their hesitation, with the open-text responses echoing the same, in the form of “missing GST filings”, “audited accounts”, or “three years of income tax filings”. The same goes for collateral, with 32 out of our 42 respondents having less than ₹25 lakh in fixed assets, and gold, vehicles, and property being the most common forms of security requested by lenders. It shows the creditworthiness gap from both sides: many respondents who have a viable business do not have the papers in order, while many lenders restrict exposure to collateral.

The gap in creditworthiness is indicated by the size of borrowing requests, too. Almost half of the respondents requested between ₹50,000 and ₹2 lakh, and most of them were aimed at restocking. Both figures suggest that smaller enterprises and lenders would benefit most from increased access to credit, as these sums are more burdensome to both borrowers and lenders compared to the typical loan size.

The secondary data section indicates areas that would benefit most from support, and the direction in which credit accessibility for MSMEs should develop. The highest demand is for subsidies and grants (22%) and less paperwork (19%) – but the next desirables also indicate actionable points for increasing credit access. Awareness of existing schemes is desired by 14% of respondents, lower interest rates for 13%, and digital lending for 13%. Two issues tie these points together: lack of support for MSMEs is lowest among respondents who have already received government aid, at 12 out of 42. This suggests that awareness and availability of existing support are at least as important as demand for new initiatives. The desirability of digital lending speaks directly to the creditworthiness issue: businesses with digital banking and payment activity are more likely to be assessed for income and expenses based on cash-flow rather than documentation. If a similar approach to lending is developed, it would reduce barriers posed by paperwork and collateral. It is important to note that those two factors are serious issues mainly for small-scale enterprises, which lack infrastructure to maintain regular banking and formal lending activity.

A few suggestions stem from these points, but they remain broad in direction only, as more research is needed into MSMEs and their credit needs. Smaller working-capital loans, designed specifically for enterprises that require them, would benefit from being simpler and quicker to arrange. The same goes for cashflow-based lending products that digitalized businesses could use maintaining digital expenditure and receivables records would allow more accurate assessment of an enterprise’s ability to repay a loan. Formalizing existing government schemes and extending outreach to smaller enterprises would improve awareness, while lessening the documentation burden for micro-enterprises with limited history of formal activity is crucial for enabling credit access. The analysis suggests the informal sector’s reluctance to engage with formal credit has opportunity costs in enterprise development and job creation, and these costs are too high to ignore.

CONCLUSION

Even though Micro Small and Medium Enterprises (MSMEs) make a substantial contribution to GDP, manufacturing, exports, employment figures, and local economy. These enterprises face several challenges in terms of accessing sufficient and affordable credit. Moreover, apart from the availability of credit resources, it is vital to pay attention to issues concerning the readiness of a business to meet the requirements of a formal lender. The research highlights the gap between the estimated credit needs of MSMEs and the supply of credit resources from conventional lenders. According to the study's calculations, the addressable debt (the amount of debt that an acquirer expects to undertake) reaches 30 lakh crore, which means that the growth of formal credit is insufficient to cover MSME's requirements. The researchers identified several factors that impact credit accessibility in MSMEs, namely, a lack of collateral, financial statements, and credit history, information asymmetry, absence of formalization, and slow response from formal lenders. Since MSMEs frequently operate on a small scale and have seasonal cash flows, these factors significantly constrain their access to credit resources. Consequently, enterprises are either unable to obtain the required amount of credit or turn to informal lenders who provide expensive and unreliable loans. The research highlights that micro enterprises are particularly dependent on informal financing sources compared to their small and medium-sized counterparts.

Furthermore, the comparison of the case study enterprises demonstrates the value of credit opportunities for MSMEs. Indeed, the firm that did not have access to formal credit had only 36-39% of its production capacity utilized, whereas the well-financed enterprise reached 87% of its capacity utilization. The latter had more revenue and created substantially more employment opportunities both directly and indirectly. Although the cases above do not establish a casual relationship between credit availability and economic performance, they demonstrate the potential benefits of sufficient financing.

The study reveals an apparent paradox of digital lending to MSMEs. Since MSMEs engage in active digital transactions, the research suggests that it is necessary to capitalize on this trend to improve credit accessibility. Thus, India should leverage the data generated by enterprises and individuals and apply it to credit assessment. Fortunately, the Indian government has taken proactive measures to support MSMEs through credit availability. CGTMSE, Udyam portal and digital credit infrastructure promote MSME financing and contribute to their overall development. Nonetheless, the fact that MSMEs still have a credit gap highlights the insufficiency of the current measures. Thus, policymakers should develop favourable conditions for enterprises to ensure timely access to credit on reasonable terms.

The research concludes that sufficient credit accessibility is essential for the continuous growth of MSMEs in terms of production, employment, and economic development. The transition from collateral-based lending to cashflow based and reliable data-based lending would reduce information asymmetry and promote MSME development. However, the study emphasizes that credit accessibility should be supported not only by technology but by an increase in financial literacy and effective implementation mechanisms. Ultimately, bridging the MSME credit gap would require the combined effort of the government, banks, NBFCs, Fintechs, and MSMEs themselves.

RECOMMENDATION

1. Improve implementation of credit guarantee schemes
 - Government credit guarantee schemes such as CGTMSE should be made easier for eligible borrowers to access.
 - Banks should be encouraged to make greater use of available guaranteed mechanisms. Bank-level implementation should be evaluated rather than measuring success only through total guarantees approved.
3. Reduce loan processing time
 - MSME loan applications should be processed within clearly defined time limits.
 - Digital document verification and automated preliminary assessment can reduce unnecessary branch visits. Faster processing is particularly important for working-capital loans because business opportunities and inventory requirements are often time-sensitive.
4. Reduce unnecessary collateral requirements
 - Eligible small-ticket MSME loans should not be unnecessarily delayed because of collateral requirements.
 - Banks should clearly communicate the security and guarantee requirements to applicants.
 - Alternative methods of assessing creditworthiness should be encouraged for businesses without substantial property assets.
5. Strengthen MSME financial literacy
 - Entrepreneurs should be provided with simple information about available government schemes, eligibility conditions, interest rates and application procedures.
 - Financial-literacy programmes should explain the benefits of maintaining proper business accounts, GST records and digital transaction histories.
6. Strengthen the role of NBFCs and FinTech lenders
 - NBFCs and FinTech platforms can complement traditional banks by serving borrowers who may not fit conventional lending models.
 - Their lending practices should continue to use responsible and transparent credit assessment.
 - Partnerships between banks, NBFCs and digital platforms can help expand credit availability while maintaining appropriate risk controls.
7. Improve formalization of micro-enterprises
 - More businesses should be encouraged to register through the Udyam system.
 - Formalization should be connected to practical benefits such as easier access to finance and government support.
 - Registration alone should not be treated as the final objective; it should help create a usable financial and business record for the enterprise.
8. Monitor the actual impact of credit
 - Government and financial institutions should evaluate whether loans result in improved business outcomes.
 - Monitoring should include production, capacity utilization, revenue, employment generation and business survival.

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Datasets & Links

- **Reserve Bank of India – Database on Indian Economy (DBIE):**
 - Link: <https://dbie.rbi.org.in/>
 - Finding historical time-series data on bank credit deployment to the priority sector, specifically breaking down credit given to Micro, Small, and medium enterprises.
- **Ministry of MSME – Udyam Registration Portal & Dashboard:**
 - Link: <https://udyamregistration.gov.in/> and <https://dashboard.msme.gov.in/>
 - Pulling the exact number of formalized/registered MSMEs categorized by state, gender of ownership, and enterprise size (Micro vs. Small) to juxtapose against the credit data.
- **Credit Guarantee Fund Trust for Micro and Small Enterprises (CGTMSE) Data:**
 - Link: <https://www.cgtmse.in/>
 - Checking the state-wise and bank-wise guarantees approved versus the actual guarantees invoked, which helps prove the "implementation bottleneck" point in the report.
- **Survey Data:**
 - Link: <https://forms.cloud.microsoft/r/rkNh7m90kl>
 - Data collected directly from business owners—covering their operations, credit histories, and loan experiences—did not provide sufficient evidence to assess the macro-level impact.